Task requirements review and implementation
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User

@README.md this is a file with task requirements. Check it on consistency, think about implementation, ask detailes if needed. Let's make everything step-by-step, after my approval

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Cursor

Reviewing the task requirements and current codebase to assess consistency and outline an implementation plan.

I read the README, `html2json.js`, and `index.html`. Here is a structured review and a proposed step-by-step plan. Nothing will be implemented until you approve each step.

README consistency check

Issues / gaps in the README itself

Issue	Details
Broken internal link	Line 31 references `[Expected Deliverables](#expected-deliverables)`, but that section does not exist. Only **Expected repository structure** is present.
Typo	Line 8: *"Is is required"* → should be *"It is required"*.
Title mismatch	README: *"Software Development Intern"* vs `index.html`: *"Software Developer Intern"*.
No JSON schema defined	Intentional — you must choose a structure and defend it at review.
No test criteria	No automated tests, no golden JSON examples, no list of required HTML features.
"DOM parser" scope is vague	Clearly forbids browser DOM parsing (`DOMParser`, `innerHTML`, etc.). Unclear whether a third-party streaming parser (e.g. htmlparser2) is allowed. Given a single `html2json.js` file and no `package.json`, a **hand-written parser** is the safest assumption.

What is clear and consistent

- Deliverable: `html2json(htmlText: string) → object`
- Must not crash on varied / edge-case HTML
- No DOM parser
- AI usage is **required**; submit conversation link in `ai_help/`
- Required folders/files: `html_samples/`, `ai_help/`, keep `index.html`
- Quality and robustness matter more than speed

Current repo state

```
html2json.js     ← stub only
index.html       ← demo UI (unchanged)
README.md
```

Missing: `html_samples/`, `ai_help/`

The built-in examples in `html2json.js` are hints, not expected output:

```56:61:html2json.js
  const jsonContent = {
    "Comment 1":
      "You have to think about how to take into account various html inputs so your json structure will cover them all and handle different cases.",
    "Comment 2":
      "When you make any choice in terms of selecting specific json structure for conversion - be ready to provide reasoning behind such choice.",
```

Implementation thinking (high level)

Core constraint: no DOM parser

You need a **custom HTML tokenizer + tree builder** in plain JavaScript inside `html2json.js`.

Typical pipeline:

```
HTML string
  → tokenize (tags, attributes, text, comments)
  → build tree (handle nesting, void tags, unclosed tags)
  → serialize to JSON object
```

Recommended JSON schema (proposed — needs your approval)

A DOM-like AST is the most defensible choice:

```json
{
  "type": "document",
  "children": [
    {
      "type": "element",
      "tagName": "div",
      "attributes": { "class": "foo" },
      "children": [
        { "type": "text", "content": "Hello" },
        { "type": "element", "tagName": "p", "attributes": {}, "children": [...] }
```

```
      ]
    }
  ]
}
```

Why this shape:
- Maps cleanly to HTML concepts (elements, text, attributes)
- Handles nesting, fragments, and multiple roots
- Easy to explain in a code review
- Common pattern (similar to htmlparser2 / DOM JSON serializers)

Alternatives (simpler but weaker): tag-name-as-key objects, flat arrays — harder to justify for complex HTML.

Edge cases to plan for (must not crash)

Category	Examples
Structure	Full documents, fragments, deep nesting, sibling elements
Void / self-closing	` `, `<img>`, `<meta>`, `<input>`, `<link>`, `<hr>`
Raw-text elements	`<script>`, `<style>`, `<textarea>` — contents not parsed as HTML
Entities	`©`, `&`, `©`, numeric entities
Attributes	Single/double quotes, unquoted, boolean attrs, `=` in values
Comments / doctype	`<!-- -->`, `<!DOCTYPE html>`
Malformed HTML	Unclosed tags, mismatched tags, stray `<`
Whitespace	Between tags, inside `<pre>`, leading/trailing text
Size	Very long text (Example 2 in the demo)
Empty / invalid input	`""`, whitespace only, non-HTML strings

Robustness rule: never throw — return a best-effort tree and optionally attach non-fatal parse notes if you want transparency.

Proposed step-by-step plan (awaiting your approval)

Step 0 — Decisions (before coding)
Lock down:
1. JSON schema (AST vs alternative)
2. Whitespace policy (preserve all vs collapse between tags)
3. Malformed HTML policy (best-effort vs error nodes)
4. Whether comments / doctype appear in output
5. Whether entities are decoded in text and attributes

Step 1 — Parser skeleton
- Tokenizer for `<`, `>`, tags, attributes
- Basic open/close tag handling
- Unit-test mentally with `<div><p>hi</p></div>`

Step 2 — Tree builder
- Stack-based nesting
- Void elements list
- Text node collection

Step 3 — Special elements
- `script`, `style`, `textarea` as raw text
- Comments and doctype as node types (if approved)

Step 4 — Robustness pass
- Unclosed tags, junk input, empty input
- Entity decoding
- Large input (Example 2)

Step 5 — Samples & deliverables
- Create `html_samples/` with varied test HTML files
- Wire `showExample1` / `showExample2` to real conversion (optional but useful)
- Create `ai_help/chatgpt_chat.txt` (or Cursor conversation link)
- Optional README update with schema rationale

Step 6 — Final manual QA
- Run all samples through `index.html`
- Confirm no crashes
- Verify deliverable checklist

Questions for you (please answer before Step 0)

1. **JSON schema** — Is the DOM-like AST above OK, or do you prefer another shape?

2. **Whitespace** — Preserve all text (including between tags), or collapse insignificant whitespace?

3. **Malformed HTML** — Best-effort tree only, or include parse warnings in the JSON (e.g. `{ "type": "error", "message": "..." }`)?

4. **Comments & doctype** — Include as nodes (`type: "comment"`, `type: "doctype"`) or skip?

5. **Entity decoding** — Decode `©` → `©` in output, or keep raw `©`?

6. **Third-party parser** — Are you OK with a from-scratch parser in one file (~150–300 lines), or do you want to ask Jito whether libraries like htmlparser2 are allowed?

7. **README update** — Do you want to add a short “Design decisions” section to README, or keep reasoning for the interview only?

8. **AI help folder** — Will you submit this Cursor chat link, or also use ChatGPT separately (README mentions `chatgpt_chat.txt` specifically)?

Once you answer these and approve **Step 0**, we can move to Step 1 and implement the parser piece by piece. I’m in **Ask mode** right now — when you’re ready for actual coding, switch to **Agent mode** and say which step to start with.

User

What is the common way to do JSON shape in tasks like that and why? What is the common behaviour of html to json parser? What kind of inputs is valid and what are not?

Cursor

Common JSON shapes (and why)

There is no single official “HTML → JSON” standard, but in tasks like yours the **DOM-like AST tree** is the most common choice.

1. DOM-like AST (most common in coding tasks)

```json
{
  "type": "element",
  "tagName": "div",
  "attributes": { "class": "container" },
  "children": [
    { "type": "text", "content": "Hello " },
    {
      "type": "element",
      "tagName": "span",
      "attributes": {},
      "children": [{ "type": "text", "content": "world" }]
    }
  ]
}
```

Why it’s popular:
- Mirrors how HTML is actually structured (tree, not flat key/value).
- Handles **nesting**, **siblings**, **repeated tags** (`<li>` × N), and **mixed content** (text + elements).
- Easy to explain in an interview: “each node has a type, optional tag/attributes, optional children.”
- Same idea as real parsers (`htmlparser2`, `parse5`, browser DOM serialization).

Node types people usually include: `element`, `text`, sometimes `comment`, `doctype`, rarely `cdata`.

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2. Tag-as-key object (simpler, weaker)

```json
{
  "div": {
    "@attributes": { "class": "foo" },
```

```
    "p": "Hello",
    "span": "world"
  }
}
```

Problems: duplicate tags collapse (`<li><li>`), order is lost, text vs child elements get ambiguous. Fine for toy examples, hard to defend for real HTML.

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3. Flat / path-based (uncommon for this task)

```json
[
  { "path": "html.body.div[0].p", "text": "Hello" }
]
```

Useful for search/indexing, not for a general “convert HTML to JSON” intern task.

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4. Library-native output (if you were allowed libs)

Tools like **htmlparser2** or **xml2js** have their own shapes. Interviewers often still want **your** schema + reasoning, not “whatever the library returns.”

Practical recommendation for Jito: DOM-like AST with `type`, `tagName`, `attributes`, `children`, `content` (for text). That’s the shape reviewers expect and the easiest to justify.

Common behavior of HTML → JSON parsers

Behavior falls between **strict XML-style** and **browser HTML5 lenient** parsing. For a hand-rolled task parser, people usually implement a **practical subset** of HTML5 rules.

Typical pipeline

```
string → tokenize → build tree → JSON
```

Stage	What happens
Tokenize	Split into tags, attributes, text, comments
Build tree	Stack of open elements; push on open tag, pop on close
Serialize	Walk tree → JSON nodes

Behaviors reviewers often expect

| Topic | Common behavior |

|-------|-----------------|
Void elements	` `, `<img>`, `<input>`, `<meta>`, `<link>`, `<hr>`, etc. never expect a closing tag; no children (except special cases)
Raw-text elements	Inside `<script>`, `<style>`, `<textarea>` (and often `<title>`), content is **literal text**, not parsed as HTML
Self-closing syntax	`<div/>` often treated as `<div>` (HTML5), not XML-empty
Case	Tag names lowercased (HTML5); attributes often lowercased too
Entities	`&`, `©`, `©` decoded in text/attributes (at least common named + numeric)
Whitespace	Often preserved in text nodes; “insignificant” space between tags may be kept or trimmed — **you pick one and document it**
Comments	Usually included as `{ type: "comment", content: "..." }` or skipped — again, document the choice
DOCTYPE	Often one node: `{ type: "doctype", name: "html" }` or ignored
Malformed HTML	**Must not crash** — best-effort tree: auto-close, ignore bad tags, wrap fragments
Fragments	`<div><p>x</p></div>` without `<html>` is valid input; multiple roots → array or synthetic `document` wrapper
Empty input	Return `{ type: "document", children: [] }` or similar, not throw

What “good” parsers avoid

- Throwing on unclosed tags
- Parsing `<script>alert('<div>')</script>` as nested `<div>`
- Treating every `<` as a tag start inside text (need proper state machine)

Valid vs invalid inputs

Important distinction: **“valid HTML5”** (spec) vs **“what your parser should accept without crashing.”**

Usually treated as valid (handle gracefully)

Input	Notes
Full document	`<!DOCTYPE html><html>...</html>`
Fragment	`<div><p>hi</p></div>` only
Void/self-closing tags	` `, `<img src="x">`, `<input type="text">`
Nested structure	headers, nav, lists, tables
Attributes	quoted, unquoted (HTML allows), boolean (`disabled`), many attrs
Entities	` `, ` `, `©`
Comments	`<!-- ... -->`
Large text	long strings in `<textarea>`, `<p>`, etc.
Mixed content	text + elements interleaved
Malformed but common	unclosed tags, wrong close order, extra `>` — **recover, don’t crash**

HTML5 spec explicitly allows **error recovery**; browsers never crash on bad HTML.

Spec-invalid but still “real world” (should not crash)

- `<p><div>...</div></p>` (invalid nesting — browsers fix it)
- `<unclosed>` without `</unclosed>`
- `<div></span>` mismatched close
- `<div class=noquotes>` unquoted attributes
- Stray `<` in text (if not part of entity)
- Empty tags everywhere
- Only whitespace
- Only comments

Your task says: *“valid HTML but still crashed or produced incorrect results”* — so they will test **edge cases that look fine** and **messy but parseable** input.

Often considered out of scope / “invalid” (still must not crash)

Input	Typical handling
Empty string `""`	Empty document / empty children
Non-HTML garbage	`"hello world"`, `{not html}`	Treat as text node or ignore junk
`null` / `undefined`	Not valid if API is `string` — coerce to `""` or return empty tree
Binary / non-string	Out of scope in browser task

Rule for this task: “invalid” does **not** mean “throw an error.” It means “not perfect HTML5” — you still return **something stable**.

Strictly invalid in XML sense (HTML parsers often still accept)

- Unescaped `<` in attribute values (rare, messy)
- Multiple `<html>` roots in one string (browsers merge/fix)
- Tags inside comments (comments are opaque)

What Jito-like tasks usually evaluate

They care less about matching Chrome’s DOM exactly and more about:

1. **Sensible JSON schema** you can defend
2. **No crashes** on samples + hidden tests
3. **Coverage**: nesting, attrs, void tags, script/style/textarea, entities, comments, fragments, long text
4. **Clear decisions**: whitespace, comments, malformed HTML policy
5. **No DOM parser** — your own tokenizer/tree builder

Short answers to your three questions

Question	Answer
Common JSON shape?	DOM-like tree (`type`, `tagName`, `attributes`, `children` / `content`)
Why?	Represents HTML faithfully; handles nesting, repeats, mixed content; easy to explain

Common parser behavior?	Tokenize → stack-based tree → lenient recovery; void tags; raw-text elements; entity decode; never throw
Valid inputs?	Documents, fragments, attrs, entities, comments, malformed-but-common HTML
Invalid inputs?	Garbage strings, empty input — still return safe JSON, don't crash

If you want to proceed step-by-step, the next decision is locking your JSON schema and three policies: **whitespace**, **comments/doctype**, and **malformed HTML recovery**. Once you pick those, Step 1 is the tokenizer design.

User

"Whitespace — Preserve all text (including between tags), or collapse insignificant whitespace?" - what is the best practice?

Cursor

Best practice: preserve text nodes, trim "insignificant" inter-tag whitespace

For an HTML → JSON converter (especially in a task like yours), the most defensible approach is:

Keep all meaningful text; drop whitespace-only text nodes that sit only between tags.

That's how most serious HTML parsers and DOM serializers behave in practice.

Why not "preserve absolutely everything"?

HTML treats whitespace **between block-level tags** as usually unimportant for rendering:

```html
<div>
  <p>Hello</p>
</div>
```

If you preserve every character, JSON often gets noisy:

```json
{ "type": "text", "content": "\n  " }
{ "type": "element", "tagName": "p", ... }
{ "type": "text", "content": "\n" }
```

That's faithful to the source string, but:
- Harder to read in tests/demo
- Often not what people mean by "content"

- Reviewers may consider it low-quality output unless you explain it

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Why not “collapse all whitespace everywhere”?

That loses real data:

```html
<p>Hello   world</p>
<pre>  indented</pre>
<textarea>
  line1
  line2
</textarea>
```

Collapsing spaces inside `<p>`, `<pre>`, or `<textarea>` is wrong. So **never globally collapse**.

Practical rule (recommended)

Situation	Behavior
Text inside elements (`<p>`, `<span>`, etc.)	**Preserve as-is** (spaces, newlines, tabs)
Whitespace-only between tags	**Usually omit** (don’t create a text node)
`<pre>`, `<textarea>`, `<script>`, `<style>`	**Preserve everything** literally
Leading/trailing space in mixed content	**Preserve** (can matter: `"Hello <em>world</em>"`)
Empty / whitespace-only input	Return empty tree, don’t crash

Pseudo-logic:

```
if text is only whitespace AND parent is "normal" element AND not in pre/textarea/script/style:
  skip text node
else:
  keep text node with exact content
```

What counts as “insignificant”?

Rough HTML5-style intuition:

- **Insignificant:** `\n`, `\t`, spaces **between** tags in normal flow
  `</div>\n  <p>` → no text node for `\n  `
- **Significant:** any text that sits next to element content or inline markup
  `Hello <b>world</b>` → keep `"Hello "`
  `<span> a </span>` → keep `" a "` if you’re being conservative (safer for a test task)

When unsure, **prefer keeping text** over dropping it. Omitting only whitespace-only nodes between tags is the sweet spot.

Best practice for your Jito task

Recommended policy:

1. **Preserve all non-whitespace-only text exactly.**
2. **Skip text nodes that are whitespace-only and appear only between element tags** (normal HTML mode).
3. **Never skip whitespace inside** `pre`, `textarea`, `script`, `style`.
4. **Document this in README** in one short paragraph — reviewers like explicit decisions.

Example:

```html
<div>  <p>Hi</p>  </div>
```

Good output:

```json
{
  "type": "element",
  "tagName": "div",
  "children": [
    { "type": "element", "tagName": "p", "children": [
      { "type": "text", "content": "Hi" }
    ]}
  ]
}
```

Not:

```json
{ "type": "text", "content": "  " }
```

One-line answer

Best practice: preserve meaningful text faithfully, but **omit insignificant inter-tag whitespace-only nodes**; always preserve whitespace in `pre`, `textarea`, `script`, and `style`.

If you want maximum safety for hidden tests, use the conservative variant: only strip whitespace-only text when it's **pure** `\n`/space/tab between two tags, and keep everything else.

Want to lock this as your Step 0 policy before we design the tokenizer?

User

Ok, let's keep it in mind for future implementation. And now let's clarify "Malformed HTML — Best-effort tree only, or include parse warnings in the JSON (e.g. { "type": "error", "message": "..." })?" - what is the best practice?

Cursor

Best practice: best-effort tree only (no error nodes in the main tree)

For a task like yours, the standard approach is:

Always return a stable JSON tree; never throw; do not mix parse warnings into the element tree.

Malformed HTML is repaired silently (auto-close tags, ignore bad markup, wrap fragments), same idea as browsers and HTML5 parsing.

Why not put `{ "type": "error", ... }` in the tree?

Issue	Why it matters
Pollutes output	Consumers expect `element` / `text` nodes, not diagnostics mixed with content
Not a standard HTML→JSON shape	DOM-like ASTs represent *document structure*, not parse logs
Hard to test	Reviewers compare structure; error nodes make golden output unpredictable
Scope creep	Full error reporting needs a real parser spec (what counts as an “error”?)

Error/warning nodes are useful in **compilers** or **linters**, not in a general HTML→JSON converter for an intern task.

What “best-effort” usually means

Malformation	Typical recovery
Unclosed tag	Auto-close when parent closes or at EOF
Wrong close tag	Close open tags until match, or ignore stray `</tag>`
Unknown / broken tag	Skip tag or treat `<` as text if clearly not a tag
Fragment (no `<html>`)	Fine — parse as-is or wrap in `document`
Garbage before first tag	Text node or skip
Empty input	`{ type: "document", children: [] }`

Rule for Jito: *must not crash* matters more than perfect HTML5 compliance.

Optional: warnings outside the tree (usually skip for this task)

Some production parsers expose:

```js
{ tree: {...}, warnings: ["Unclosed <div> at line 5"] }
```

That's fine for a library API, but your contract is:

```js
html2json(htmlText) → object
```

So a **single tree object** is the right fit. Adding a top-level `{ document, warnings }` changes the API and isn't asked for in the README.

If you want extra credit without breaking the demo UI, you could document recovery rules in README only — not in JSON.

What reviewers likely care about

1. **No throws** on messy input
2. **Sensible structure** for common cases (`<div><p>x</p>`, unclosed `<div>`, etc.)
3. **Predictable rules** you can explain in the interview

They're less likely to check whether you emitted `"type": "error"` for a mismatched `</span>`.

Recommended policy for your implementation

```
Malformed HTML → silent recovery → normal element/text tree
Never throw
No error/warning nodes in JSON
Optional: describe recovery rules in README ("unclosed tags are auto-closed")
```

One-line answer: Best practice is **best-effort tree only** — repair malformed HTML quietly and return a clean DOM-like JSON structure; save explicit error reporting for linters, not for this converter.

When you're ready for Step 0, we'll have two policies locked:

- Whitespace: omit insignificant inter-tag whitespace-only nodes

- Malformed HTML: silent recovery, no error nodes

Still open: **comments/doctype** and **entity decoding** — want those clarified next?

User

I think it would be better add warnings. Keep it in mind

Cursor

Good call — warnings can work well if you structure them cleanly. Here's a practical way to do it without breaking the demo or making output hard to review.

Recommended shape with warnings

Keep the **tree as the main payload**, and put warnings **alongside** it, not mixed into `children`:

```json
{
  "type": "document",
  "children": [ ... ],
  "warnings": [
    { "message": "Unclosed tag <div>", "position": 42 },
    { "message": "Unexpected closing tag </span>", "position": 108 }
  ]
}
```

Or, if you prefer a flatter API:

```json
{
  "document": { "type": "document", "children": [ ... ] },
  "warnings": [ ... ]
}
```

Avoid `{ "type": "error" }` nodes inside `children` — that confuses structure with diagnostics.

Why this fits your task

Benefit	Detail
Shows edge-case awareness	Reviewers see you noticed malformed input
Doesn't break parsing	Tree is still usable even when warnings exist
Easy to explain	"Structure in `children`, diagnostics in `warnings`"
Demo still works	`JSON.stringify` in `index.html` shows both

The demo calls `html2json()` and stringifies the whole object — a top-level `warnings` array will show up automatically.

What to warn about (keep the list small)

Stick to **high-signal, recoverable** cases:

- Unclosed tag at EOF
- Unexpected/mismatched closing tag
- Invalid or incomplete tag syntax (recovered by treating as text)
- Unknown void element misuse (optional)

Skip noisy warnings for every tiny HTML5 quirk — 1–3 clear messages per bad input is enough.

Warning object fields (minimal)

```json
{
  "message": "Unclosed tag <section>",
  "position": 156
}
```

`position` = character index in the input string (cheap to implement, useful for debugging). Line/column is optional.

Updated Step 0 policies (for when we implement)

Policy	Decision
JSON schema	DOM-like AST (`document` → `element` / `text` / …)
Whitespace	Omit insignificant inter-tag whitespace-only text nodes
Malformed HTML	Best-effort tree + **`warnings` array** (sibling to tree, not inside `children`)
Must not crash	Always return an object; empty input → empty tree + no warnings

Still to decide before coding: **comments/doctype** (include or skip?) and **entity decoding** (`©` → `©` or keep raw?).

When you want to lock those two and move to Step 1 (tokenizer design), say the word.

User

Include comments and doctype.

And what is the best practice for entity decoding?

Cursor

Comments & doctype — noted

Locked for implementation:

```json
{ "type": "doctype", "name": "html" }
{ "type": "comment", "content": " sidebar " }
```

Usually as siblings under `document.children`, in source order (before `<html>` when present).

Entity decoding — best practice

Decode entities in text nodes and attribute values; keep tag names raw (already plain ASCII).

That matches browsers and most HTML→JSON tools: JSON holds the **logical** content, not the HTML-encoded transport form.

Decode

Form	Example	Output
Named	`©`	`©`
Named	`&`	`&`
Named	`<`	`<`
Decimal	`©`	`©`
Hex	`©`	`©`
Common named set	` `, `"`, `'`	respective chars

Apply in:
- `{ "type": "text", "content": "..." }`
- `attributes` values, e.g. `"title": "Tom & Jerry"`

Do **not** decode

Context	Why
Inside `<script>`, `<style>`	Literal source; `<` may be JS/CSS, not HTML entities
Tag names / attribute names	Never entity-encoded in valid HTML
Comments	Opaque — `<!-- © -->` stays `©`
DOCTYPE	Literal string
Unknown/invalid entities	Leave as-is (see below)

Your Example 1 has `©` in `<footer>` — decoded text `"© 2024 My Website"` is the expected, reviewer-friendly output.

How deep to go (practical scope)

For this task, **HTML5 “safe” subset** is enough:

1. **5 essential named:** `&`, `<`, `>`, `"`, `'`
2. **Common extras:** ` `, `©`, `®`, `™`, etc. (20–30 frequent ones)
3. **Numeric:** `&#DDDD;` and `&#xHHHH;` (with optional `;`)

Invalid / unknown entities: don’t crash; leave unchanged:

- `&copy` (no semicolon) → often still decode as `©` in HTML5; document if you do
- `&unknown;` → keep `&unknown;`
- `&` at end of string → keep `&`

Conservative rule: **only decode when you're confident**; otherwise preserve the original substring.

Order of decoding (important)

Decode **numeric and named entities**, with **`&` handled first** in a multi-pass or careful single pass:

```
&amp;copy;  →  &copy;  →  ©   (if you decode &amp; first, then &copy;)
```

Or one pass that resolves full entity tokens correctly. Wrong order causes double-decode bugs.

Recommended policy for your task

Location	Entity decoding
Text in normal elements	**Yes**
Attribute values	**Yes**
`pre`, `textarea`	**Yes** (still text content)
`script`, `style`	**No** (raw literal)
Comments	**No**
DOCTYPE	**No**

Updated Step 0 checklist

| Policy | Decision |

|--------|----------|
JSON schema	DOM-like AST
Whitespace	Skip insignificant inter-tag whitespace-only nodes
Malformed HTML	Best-effort tree + `warnings[]`
Comments	**Include** (`type: "comment"`)
DOCTYPE	**Include** (`type: "doctype"`)
Entities	**Decode in text + attributes**; skip in script/style/comments/doctype

That’s a complete, defensible spec. Next step when you’re ready: **Step 1 — tokenizer design** (states, token types, void-element list). Switch to Agent mode when you want implementation to start.

User

Let's create project structure first

Cursor

Creating the required project folders and starter files from the README.

Creating `html_samples/` and `ai_help/` with starter sample files aligned to our agreed test categories.

Project structure is in place. Current layout:

```
jito-intern-test-task/
├── html2json.js        (stub — unchanged)
├── index.html          (demo UI — unchanged)
├── README.md           (unchanged)
├── html_samples/
│   ├── 01-full-document.html      ← built-in Example 1
│   ├── 02-fragment-long-text.html ← built-in Example 2
│   ├── 03-void-elements.html      ← img, br, hr, input, meta, link
│   ├── 04-entities-and-attributes.html
│   ├── 05-comments-and-doctype.html
│   ├── 06-malformed.html          ← unclosed/mismatched tags
│   ├── 07-raw-text-elements.html  ← script, style, textarea, pre
│   └── 08-edge-cases.html         ← whitespace-only input
└── ai_help/
    ├── chatgpt_chat.txt           ← required by README (marked N/A)
```

```
    └─── cursor_conversation.txt     ← placeholder for your Cursor chat link
```

Notes:
- Sample files map to the categories we agreed on (doctype, comments, entities, malformed + warnings, etc.).
- `ai_help/chatgpt_chat.txt` points to `cursor_conversation.txt` since you're using Cursor, not ChatGPT.
- Before submission, paste your shareable Cursor conversation link into `cursor_conversation.txt` and verify it opens in incognito.

Next step (when you approve): Step 1 — implement the tokenizer/parser skeleton in `html2json.js` using our locked policies:

- DOM-like AST with `document`, `element`, `text`, `comment`, `doctype`
- Whitespace: skip insignificant inter-tag whitespace-only nodes
- Malformed HTML: best-effort tree + `warnings[]`
- Entity decoding in text/attributes (not in script/style/comments)

Ready to start Step 1?

User

Let's create a detailed plan of project implementation, every task sould be well written. Group related tasks in one general point and write detailed subpoints for each of them. I also want to add separate point where would be implementation of html2json function itself. Keep in mind, I want to use TDD, keep main functional programming principles (like pure functions, immutability etc.) and keep DRY, KISS, YAGNI principles also

---

Cursor

Reviewing the project state and agreed design decisions to draft a detailed TDD implementation plan.

Checking for an misplaced test file, then drafting the plan from our agreed design decisions.

Test setup preference

> **For TDD, which test runner do you prefer? (No package.json exists yet.)**
> - Node.js built-in test runner (node:test) — zero extra deps, KISS/YAGNI ✔
> - Vitest — faster DX, more features
> - Jest — familiar API

User

Add our JSON shema before html2json function in @html2json.js

---

Cursor

Adding the agreed JSON schema as a block comment before `html2json`.

Added the JSON schema as a block comment directly above `html2json` in `html2json.js`. It documents:

- **Root** — `{ type: "document", children, warnings }`
- **Node types** — `element`, `text`, `comment`, `doctype` with their fields
- **Warning shape** — `{ message, position }`
- **Policies** — whitespace, malformed HTML, entity decoding rules we agreed on

The old stub instruction comment was replaced by this schema. Ready for the next step when you are.

User

Inspect my implementation, which tests should we also add to cover all possible cases? Implement all only after my approval

---

Cursor

I'll review your current parser and existing tests, then propose a focused "missing cases" test list before we implement anything.

I found your current parser and test suite; next I'll run tests once and map uncovered edge cases against your target schema/policies.

Great progress: your current tests all pass (`15/15`), and they cover basic structure, attributes, void/self-closing tags well.

What's still missing is coverage for the **contract we agreed** (doctype/comments/warnings/entities/whitespace policy).
Right now implementation/test behavior is still centered on a simpler schema (`tag` and no `warnings`).

Gaps I found in current implementation

- Output shape mismatch with agreed schema:
  - uses `tag`, not `tagName`
  - no `warnings` field at root
- No parsing for `<!DOCTYPE ...>` and `<!-- comments -->`
- No entity decoding in text/attributes (`©`, `&`, `©`, etc.)
- Whitespace handling is currently `trim()`-based, so meaningful spaces can be lost
- Malformed HTML recovery is minimal (only pops exact top tag), no warning emission

- No special raw-text handling for `script/style/textarea` (content can be parsed incorrectly)

Tests we should add (recommended)

P0 — Must-have (core contract)
1. Returns root with `type`, `children`, **and `warnings` array**
2. Uses `tagName` for elements
3. Parses doctype node: `<!DOCTYPE html>`
4. Parses comment node: `<!-- note -->`
5. Decodes entities in text (`©`, `©`, `©`, `&`)
6. Decodes entities in attribute values (`title="Tom & Jerry"`)
7. Does **not** decode entities in comment content
8. Does **not** decode entities in `script/style` text
9. Keeps best-effort tree for malformed HTML and adds warning(s)
10. Non-string input still returns valid document object with `warnings`

P1 — Important behavior edges
11. Mismatched close tag creates warning but parser continues
12. Unclosed tags at EOF create warning(s)
13. Stray closing tag (`</span>`) creates warning
14. Multiple top-level roots remain ordered
15. Preserves significant text spaces: `<span> a </span>`
16. Drops whitespace-only inter-tag text nodes in normal elements
17. Preserves whitespace in `<pre>`
18. Preserves whitespace in `<textarea>`
19. Raw text in `<script>` containing `<div>` is kept as text, not parsed as tag
20. Raw text in `<style>` containing `{}` and `<` is kept as text

P2 — Robustness / regression
21. Attribute edge: mixed quotes + boolean + unquoted in one tag
22. Uppercase tags normalized to lowercase
23. Unknown entity remains unchanged (`&unknown;`)
24. Empty input returns empty doc with empty warnings
25. Fixture sweep: every file in `html_samples/*.html` runs without throw

Optional hardening
26. Fuzz smoke test: random noisy strings never throw (100–500 iterations)
27. Very long text node performance sanity (no crash)

If you approve, I'll implement these tests in a structured way:

- `tests/unit/*` for entities/whitespace/recovery helpers
- `tests/integration/html2json.test.js` for full parser behavior
- fixture-driven no-crash test over `html_samples/`

And only after that, we'll adapt implementation to make them pass (strict TDD).

User

I don't want overengineering and I need only nessesary tests for this task. Let's keep 'tag' instead 'tagName' i fixed the shema. And I also wanted to handle errors (add warnings) at the end of my core implementation. Is it good enough or it is better to do now during implementation?

Cursor

Yes — that's a good direction, and not overengineering.

1) `tag` vs `tagName`

Keeping `tag` is totally fine **if you keep it consistent** everywhere:

- parser output
- tests
- README/schema comment

For this task, consistency matters more than the exact key name.

2) Warnings now or later?

Better to design for warnings **now**, even if you implement full warning logic later.

Practical compromise (minimal effort, low risk):
- add `warnings: []` to root shape now
- keep parser behavior same for now
- add warning-producing rules at the end

Why: if you postpone shape change, you'll have to touch many tests/output expectations twice.

3) Minimal necessary tests (not overengineered)

You already cover many basics. For this task, I'd keep only these additions:

1. Root always has `warnings` array
2. Doctype parsing (`<!DOCTYPE html>`)
3. Comment parsing (`<!-- ... -->`)
4. One malformed HTML case returns non-empty `warnings` and doesn't crash
5. Entity decode in text (`©`, `&`, numeric)
6. Entity decode in attribute value
7. Raw text in `<script>` is not parsed as nested tags
8. Whitespace-only between tags is skipped
9. Whitespace preserved in `<textarea>` or `<pre>`
10. Fixture smoke test: all `html_samples/*.html` do not throw

That's enough to prove robustness and edge-case handling without turning this into a spec-complete parser.

Recommendation

- Keep current scope and simplicity (KISS/YAGNI).

- Introduce `warnings` field now (empty initially).
- Implement actual warning emission near the end, but before final QA.
- Add only the 10 tests above.

If you want, I can next give you a strict **ordered implementation sequence** (5–7 steps) that keeps refactoring minimal and aligns with your current code style.

User

I completed my implementation. Check whether we need more test cases or not based on requirements?

Cursor

Reviewing your implementation and tests against the README requirements and our agreed schema.

Running a quick no-crash check on `html_samples/` to compare with test coverage.

You're in good shape. **43/43 tests pass**, and manual smoke on all `html_samples/*.html` shows **no crashes**.

For Jito requirements, you likely need **only 1–2 more tests**, not a big expansion.

What is already well covered

Your suite already covers the important task areas:

- Basic tree building (text, nesting, siblings)
- Attributes (quoted/unquoted/boolean)
- Void + self-closing tags
- Comments + doctype
- Entity decoding (named, decimal, hex, malformed entity kept)
- Raw text handling (`script`, `style`, `textarea`, `title`)
- Quote-aware raw closing logic
- Malformed HTML recovery (no throw, best-effort output)
- Whitespace behavior (you chose "preserve", and tests match that)

That is enough for "various HTML structures" and "must not crash" in most reviews.

What is missing (important)

1) `html_samples` automated no-crash test — **recommended**
README explicitly says to verify your own samples before submit.
You currently have samples, but no test that reads them.

Add one test like:
- loop through `html_samples/*.html`

- call `html2json(fileContent)`
- assert it does not throw
- assert result has `type: "document"` and `Array.isArray(children)`

This is the most valuable missing test.

2) One integration smoke test for full document — **optional but nice**
Use `01-full-document.html` and assert a few key things:
- has doctype node
- has nested `header/nav/main/footer`
- decodes `©` in footer text

Not required if you add the fixture sweep, but good for confidence.

What you can skip (to avoid overengineering)

You do **not** need more tests for:

- full golden JSON for every sample file
- fuzz/random input generator
- exhaustive HTML5 spec cases
- duplicate tests for `null` and `undefined` (one invalid-input test is enough)
- warning assertions (since warnings are not implemented yet)

Implementation vs schema note (not test gap, but worth fixing)

Your schema comment mentions `warnings`, but output currently has no `warnings` field:

```362:385:html2json.js
function html2json(htmlText) {
  if (typeof htmlText !== "string") {
    return {
      type: "document",
      children: [],
    };
  }
  // ...
}
```

Also doctype uses `content`, while schema comment says `name`. Tests follow `content`, which is fine — just keep schema comment consistent.

Final recommendation

Add only this now:
1. `html_samples` no-crash fixture test

Optional:
2. one full-document integration smoke test

After that, your test set is sufficient for task requirements without overengineering.

If you want, I can draft the exact test code for those 1–2 cases (you can paste manually, or switch to Agent mode and I'll add them).

User

Apply all your tests recommendations (html_samples and full-document integration smoke test)

Cursor

Adding the `html_samples` no-crash sweep and the full-document integration smoke test.

Added two tests to tests/html2json.test.js:

1. **`does not crash on html_samples fixtures`** — loops through all `html_samples/*.html` files and asserts each returns a valid `{ type: "document", children: [...] }` without throwing.

2. **`parses full document sample with key structure and decoded entities`** — loads `01-full-document.html` and checks:
   - doctype node (`content: "html"`)
   - `<html lang="en">`
   - presence of `header`, `nav`, `main`, `footer`
   - footer text decodes to `"© 2024 My Website"`

Also added a small `findByTag` helper for tree traversal in the integration test.

Result: `45/45` tests passing.

User

How would it better to handle errors in my @html2json.js ?

Cursor

Reviewing your current error-handling approach in `html2json.js`.

Your current approach is **safe** (mostly no-throw), but errors are mostly **silent**. That's fine for basic parsing, but not ideal for the `warnings` contract in your schema comment.

What you have now

1. **Recovery without reporting** in `closeTagWithRecovery()` — mismatched closes are handled, but no warning is recorded.
2. **Fallback to text** for invalid `<...` sequences — good, but unreported.
3. **Global `try/catch` in `html2json()`** — returns empty tree on unexpected failure; tree is lost.

4. **No `warnings` field** in returned object, even though schema documents it.

```480:504:html2json.js
function html2json(htmlText) {
  if (typeof htmlText !== "string") {
    return {
      type: "document",
      children: [],
    };
  }
  // ...
  try {
    return parseHtml(trimmedHtml);
  } catch {
    return {
      type: "document",
      children: [],
    };
  }
}
```

Better approach for this task

Use **warnings, not thrown errors**.

1) Always return stable root shape

Every path should return:

```js
{
  type: "document",
  children: [],
  warnings: []
}
```

Including invalid input, empty input, and catch fallback.

2) Collect warnings during parse (best place)

Pass a `warnings` array through parser state (or return `{ tree, warnings }` from `parseHtml`).

Add warnings only for high-value cases:

Event	Example warning
Unexpected closing tag	`Unexpected closing tag </span> at line 42`
Auto-closed unclosed tags at EOF	`Unclosed tag <div> at line 10`
Mismatched close caused implicit closes	`Closing </div> implicitly closed <span>`
Unclosed comment	`Unclosed comment starting at line 5`
Unclosed raw element (`script/style/...`)	`Unclosed <script> at line 20`

Keep the list small (5–6 types max).

3) Keep `position`/`line` consistent

Your schema says `line`, comment says character index. Pick one and stick to it.
For your parser, **`position` (char index)** is easiest because you already track `index`.

```js
{ message: "Unexpected closing tag </span>", position: 108 }
```

4) Pure helper for warnings (FP-friendly)

```js
function addWarning(warnings, message, position) {
  return [...warnings, { message, position }];
}
```

Or mutate a local `warnings` array inside `parseHtml` only (pragmatic, still simple).

5) Where to emit warnings in your code

- **`closeTagWithRecovery()`**
  - if no matching open tag → warning
  - if popped more than one node → warning for implicit closes
- **End of `parseHtml()`**
  - if stack length > 1 → warn for each unclosed element
- **`tryReadComment()` / raw element readers**
  - if opening found but no closing before EOF → warning
- **Invalid `<` token fallback**
  - usually no warning (too noisy), unless you want debug-level warnings

6) Keep `try/catch` as last-resort only

Current catch drops parsed output. Better:

```js
try {
  return parseHtml(trimmedHtml);
} catch (error) {
  return {
    type: "document",
    children: [],
    warnings: [{
      message: "Internal parser error",
      position: 0
    }]
  };
}
```

You should rarely hit this if recovery is solid.

Minimal implementation plan (no overengineering)

1. Add `warnings: []` to root in `parseHtml`.
2. Implement 3 warning rules first:
   - unexpected closing tag
   - unclosed tags at EOF
   - unclosed comment/raw block
3. Always include `warnings` in `html2json` returns.
4. Add 2–3 tests:
   - malformed sample returns `warnings.length > 0`
   - valid simple HTML returns `warnings: []`
   - parser never throws on samples

That's enough for reviewers.

What not to do

- Don't throw from parser for malformed HTML.
- Don't add `{ type: "error" }` nodes inside `children`.
- Don't add dozens of warning codes/types.
- Don't build a full linter.

If you want, I can next give you a concrete patch plan mapped to your existing functions (`parseHtml`, `closeTagWithRecovery`, `tryReadComment`) function-by-function before you implement.

User

Let's implement warnings for invalid inputs, I need them to contain message and line of the problem (understandable for user, where the incorrect tag appear for example). It needs to be user friendly, so user could understand on what line he put irrelevant tag, for example. And I also want to use try...catch (error) to handle parser bugs (unexpected exeptions). Implement all after my approval

---

Cursor

Perfect request. We can implement this cleanly without overcomplicating parser logic.

Proposed implementation (no changes yet)

1) Warning shape and root contract
Return this on every `html2json()` call:

```js
{
  type: "document",
  children: [...],
  warnings: [
    { message: "Unexpected closing tag </span>", line: 12 }
  ]
}
```

- `line` is 1-based
- `warnings` always exists (empty for valid input)

2) Line tracking (user-friendly)
Add a tiny helper for line conversion from parser index:

- Precompute line-start offsets once per parse
- `getLineByIndex(index)` resolves the line number for any warning position

This keeps warnings understandable: “line where problematic token starts”.

3) Where warnings are emitted
We’ll add warnings for the most useful invalid-input cases:

1. **Unexpected closing tag** (`</x>` with no matching open tag)
2. **Implicitly closed tags** (mismatch recovery pops different tag(s))
3. **Unclosed tags at end of file** (`<div><p>...`)
4. **Unclosed comment** (`<!-- ...` without `-->`)
5. **Unclosed raw element** (`<script>`, `<style>`, `<textarea>`, `<title>` without close)

No noisy warnings for every malformed character — only actionable ones.

4) Keep try/catch for parser bugs (unexpected exceptions)
In `html2json()`:

- `try` normal parse path
- `catch (error)` returns safe document with one generic warning like:
  - `"Internal parser error. Please check your HTML and try again."`
  - line `1` (or best known line if available)

This is for parser bugs only, not regular invalid HTML.

5) Test updates
Add minimal tests:

- warnings array exists for all outputs
- valid HTML => `warnings: []`
- each warning case above emits meaningful message + correct line
- malformed sample does not crash and has warnings
- forced internal error path still returns document + warning (can be done by temporarily mocking helper in test if needed)

Implementation order
1. Add warning model + line resolver utilities
2. Wire warning collector through parser flow
3. Emit warnings in recovery points
4. Add catch-warning path for unexpected exceptions
5. Update tests to cover warnings + line numbers

If you approve, I'll implement this now in `html2json.js` and extend `tests/html2json.test.js` accordingly.

User

Let's add warning model + line resolver utilities

Cursor

Implementing this first step now: I'll add warning data model helpers and line-resolver utilities in `html2json.js` only, without wiring parser behavior yet.

Added the first warning step in `html2json.js`: **warning model + line resolver utilities**.

What was added
- `createWarning(message, line)`
- `createWarningCollection()`
- `buildLineStartIndexes(text)`
- `resolveLineNumber(lineStartIndexes, index)` (binary search)
- `createLineResolver(text)` (returns index → line resolver function)

These are utility-only additions for the next step (wiring warnings into parser flow).

Validation
- Ran full test suite: **45/45 passing**.

Next step (when you approve): wire these utilities into parse flow and start emitting user-friendly warnings for malformed input cases.

User

approve

Cursor

Great — I'll wire warnings into parser flow now (unexpected closing tags, mismatches, unclosed structures, and catch fallback), then add targeted tests for message + line.

I'm now wiring warnings into parse/recovery paths and `html2json` fallback, while keeping valid-output behavior unchanged. Then I'll adjust only the affected malformed tests and add line-number assertions.

I've wired parser warnings; now I'm updating impacted tests and adding focused line-number warning checks.

Implemented this step: warning model utilities are now **wired into parsing and recovery flow** with user-friendly line numbers, plus parser bug fallback via `try...catch (error)`.

What I changed

- Added warning emission for invalid HTML scenarios:
  - unexpected closing tag
  - implicit auto-close on mismatched nesting
  - unclosed comment
  - unclosed raw-text tag (`script/style/textarea/title`)
  - unclosed open tags at EOF
- Added line-aware warning generation using your line resolver utilities.
- Added non-string input warning in `html2json`.
- Kept `try...catch (error)` as final protection and return a safe warning on internal parser failure.
- Preserved existing valid parsing behavior.

Test updates

Updated and added tests in `tests/html2json.test.js`:

- Adjusted expected output where warnings are now emitted for malformed cases.
- Added warning-line tests:
  - unexpected closing tag reports line
  - unclosed comment reports line

Verification

- Ran full suite: **47/47 passing**
- Lints: **no issues** for edited files

If you want, next I can do the final polish step: make warning messages even more human-readable (e.g., include expected tag names in mismatch guidance).

User

I changed my warning model, I think it would be more user friendly. Check if I fix it's usage properly, if no - fix it

---

Cursor

I'll review your updated warning model and its usage across parser + tests, then fix any inconsistencies and re-run the suite.

I found the issue: warning shape was changed, but several call sites and tests still use the old single-line model. I'm fixing those mismatches now.

You were right to question it — usage wasn't fully consistent after changing the warning model, and I fixed it.

What was wrong
- `createWarning(message, openedAt, detectedAt)` was called in several places with only one line value, so `detectedAt` became `undefined`.
- Tests still expected old shape (`line`) in multiple cases.

What I fixed

In `html2json.js`:
- Updated all warning calls to pass **both** fields:
  - unclosed comment: `openedAt` + `detectedAt`
  - unclosed raw text tag: `openedAt` + `detectedAt`
  - unclosed element at EOF: both fields set to opening line
  - non-string input fallback: both `1`
  - catch fallback: both `1`

In `tests/html2json.test.js`:
- Updated warning expectations from `line` to:
  - `openedAt`
  - `detectedAt`
- Adjusted one expectation for unexpected closing tag (`openedAt: 1`, `detectedAt: 2`) to match parser behavior.

Verification
- Test suite: **47/47 passing**
- Lints: **no errors**

If you want, next I can also rename the two test titles that still say "line" so naming matches the new `openedAt/detectedAt` model.

User

ok

Cursor

Great — warning model usage is now consistent and green.

If you want, next logical steps are:
- align README/schema wording with `openedAt`/`detectedAt`,
- run one final manual check in `index.html`,
- then prepare commit/submission checklist.

User

Now analyze the project and implementation, check robustness in my solution

---

Cursor

I'll do a robustness-focused review: run current tests, inspect parser/error paths, and report concrete risks ordered by severity.

Findings (by severity)

- **High — Line numbers in warnings can be wrong because input is trimmed before parsing.**
  `html2json` calls `trim()` and then parses the trimmed string, so all line calculations are based on the modified input. If user HTML starts with blank lines/spaces, `openedAt`/`detectedAt` shift upward and point to the wrong line.
```624:635:html2json.js
  const trimmedHtml = htmlText.trim();

  if (trimmedHtml === "") {
    return {
      type: "document",
      children: [],
    };
  }

  try {
    return parseHtml(trimmedHtml);
```

- **Medium — Output shape is not stable: `warnings` is missing when there are no warnings.**
  Your schema says root always has `warnings`, but parser only adds it conditionally. This forces consumers to branch on presence and weakens API robustness.
```591:604:html2json.js
  while (stack.length > 1) {
    const unclosedElement = stack.pop();
    warnings.push(
      createWarning(
        `Unclosed tag <${unclosedElement.tag}> was automatically closed at the end of input.`,
        getOpenLine(unclosedElement),
        getOpenLine(unclosedElement)
      )
    );
  }

  if (warnings.length > 0) {
    root.warnings = warnings;
  }
```

- **Medium — `openedAt` for unexpected closing tags is semantically ambiguous.**
  For `</span>` with no opening tag, `openedAt` is taken from current stack parent (often `1` or parent line), not the problematic tag line. That may confuse users and make warning less "friendly."

```472:478:html2json.js
  if (foundTagIndex === -1) {
    warnings.push(
      createWarning(
        `Unexpected closing tag </${tag}>. No matching opening tag was found.`,
        getOpenLine(stack[stack.length - 1]),
        tokenLine
      )
```

- **Low — Internal parser error warning always reports line `1`.**
  This is safe, but less diagnostic for real bugs; if possible, include best-known `detectedAt`.

```633:644:html2json.js
  } catch (error) {
    return {
      type: "document",
      children: [],
      warnings: [
        createWarning(
          "Internal parser error occurred. Please review your HTML input and try again.",
          1,
          1
```

Open questions / assumptions

- I assume you want warning coordinates to refer to the **original user input**, not normalized input.
- I assume your new model intent is:
  - `openedAt`: line where related opening context starts (or same as detected if none)
  - `detectedAt`: line where parser detected issue

Quick robustness summary

- ✅ Strong: no-crash behavior, raw-text handling, entity decoding, broad test coverage (`47/47` passing).
- ⚠️ Main robustness gap: **line accuracy and consistency of warning payload shape**.

If you want, I can apply a minimal hardening patch next:
1) parse original `htmlText` (no trim for parsing),
2) always return `warnings: []`,
3) define/standardize `openedAt` for unexpected close tags.

User

Are that the only gaps in my project? Did you inspect the hole project?

Cursor

Good question — I'll do one quick full-repo sweep now and tell you exactly what I did and didn't inspect.

Short answer: **no, those are not the only potential gaps** — and now yes, I've done a full sweep of the current repo files.

I checked:
- parser implementation: `html2json.js`
- tests: `tests/html2json.test.js`
- task requirements: `README.md`
- demo UI: `index.html`
- support files (`html_samples`, `.gitignore`, `ai_help`)

Also ran full tests: **47/47 pass**.

Additional gaps/risk points I see

- **Warning line accuracy risk** (still the biggest): you parse `trimmedHtml`, so warning lines can shift if input starts with blank lines/spaces.
- **Inconsistent root shape**: for valid input you often return no `warnings` field; schema says root always has `warnings`.
- **`ai_help/chatgpt_chat.txt` looks incomplete**: line 4 is blank after "See the following file:". For submission clarity, this should explicitly point to your actual link/file.
- **Potential submission noise files**:
  - `ai_help/tests/html2json.test.js`
  - `ai_help/plans/html2json_tdd_implementation_f3b9c001.plan.md`

  These look accidental and may confuse reviewers about expected deliverables.
- **README internal issue** (from template): broken anchor `#expected-deliverables` and typo "Is is required" — not your fault, but if you edit README, you can clean it.

Robustness status overall

- **Core parser robustness: good** (broad malformed/edge coverage, no-crash behavior, raw-text handling).
- **Remaining robustness polish**: mostly API consistency (`warnings` always present) and line-source correctness (parse original input for diagnostics).

If you want, I can give you a strict "final pre-submit checklist" with exactly what to fix vs what to leave.

User

Let's fix warning line accuracy risk

Cursor

I'll fix this by parsing the original input (only using `trim()` for the empty-input check), so warning line numbers map to the user's real text. Then I'll run tests.

Done — fixed warning line accuracy risk.

What changed
In `html2json()`, parser now uses the **original input** for parsing and line resolution:

- still uses `trim()` only to detect empty input
- no longer parses `trimmedHtml`
- warnings now map to real user input line numbers (including leading blank lines/spaces)

Verification
- Tests: **47/47 passing**
- Lints: no issues

If you want, next I can fix the second robustness gap too: make root shape always include `warnings: []` for consistency.

User

ok

Cursor

Great.
If you want to continue, the best final hardening is:

- always return `warnings: []` on root for consistent API,
- sync test names mentioning "line" with `openedAt/detectedAt`,
- quick cleanup of extra accidental files before submission.

User

fix first two, skip file structure cleanup

Cursor

I'll make two focused changes now: enforce `warnings: []` in all root returns and rename outdated test titles that still mention `line`. Then I'll run tests.

I introduced `warnings: []` consistently, which broke many tests expecting old root shape. I'll add a small test helper shim so existing expectations remain concise while still enforcing warnings where explicitly asserted.

Done — both requested fixes are implemented.

What I changed

1. **Consistent root shape (`warnings: []`)**
   - `parseHtml()` now initializes root with `warnings: []` and always returns it.
   - empty-string return in `html2json()` now also includes `warnings: []`.

2. **Renamed outdated test titles**
   - Updated warning tests to mention `openedAt/detectedAt` instead of `line`.

Test compatibility update

Because many legacy tests expected document objects without `warnings`, I added a small test helper shim in `tests/html2json.test.js` that auto-injects `warnings: []` into expected root documents **only when expected omits warnings**.
This keeps tests concise and still strictly validates warning-heavy cases where warnings are explicitly asserted.

Verification

- Tests: **47/47 passing**
- Lints: no issues in modified files

User

Analyze logic one more time. Any addition advicese or bugs in my implementation?

Cursor

Reviewing the latest implementation for logic bugs and remaining robustness gaps.

I re-read the full parser flow (`html2json.js`) with your latest warning/line fixes. Overall it's solid for this task, but there are a few real logic gaps and some doc/demo mismatches.

Overall verdict

- **Robust for task requirements:** good coverage, no-crash behavior, raw-text handling, entity decoding, malformed recovery.
- **Not production HTML5 parser** (and that's fine), but a few edge behaviors are worth knowing before submission.

Real logic issues (worth fixing or explaining)

1) Schema says whitespace is omitted — code preserves it

Your schema comment says:

```53:55:html2json.js
  - Whitespace-only text nodes between tags are omitted (except inside pre, textarea, script, style).
```

But `flushTextBuffer` always keeps non-empty text, including whitespace-only nodes. Tests also expect preserved whitespace.

Impact: not a crash bug, but reviewers may ask why comment and behavior differ.
Advice: either update schema comment to “preserve whitespace” or implement normalization.

---

2) Unclosed comment is only warned, not treated as comment mode

For input like:

```html
<!-- note
<p>Hi</p>
```

Flow:
- `tryReadComment` fails (no `-->`)
- parser emits one warning
- then continues normal parsing, so `<p>` becomes a real element

Impact: warning says “comment not closed”, but parser still parses following tags as HTML.
Advice: acceptable for best-effort, but mention this in README/interview: “we warn, we don’t enter full comment state machine.”

---

3) Unexpected closing tag `openedAt` can point to parent, not bad tag

```472:478:html2json.js
        getOpenLine(stack[stack.length - 1]),
        tokenLine
```

For `</span>` inside open `<div>`, `openedAt` may be `<div>`’s line, not where user typed `</span>` (that’s `detectedAt`).

Impact: UX confusion, not crash.
Advice: for unexpected close, consider `openedAt = detectedAt` (or omit `openedAt`) since there is no valid opening tag.

---

4) EOF unclosed-tag warning uses same line for both fields

```592:599:html2json.js
        getOpenLine(unclosedElement),
        getOpenLine(unclosedElement)
```

`detectedAt` could be last input line (EOF), while `openedAt` stays opening line.
Advice: set `detectedAt` to last line of document for clearer diagnostics.

---

5) `<pre>` is not raw-text mode

`script/style/textarea/title` are raw; `pre` is not. So:

```html
<pre><b>x</b></pre>
```

`<b>` is parsed as an element, not literal text.
If you intended pre to preserve literal content, current logic differs.

Minor inconsistencies (low severity)

- **Attribute type in schema:** says string values, but boolean attrs are `true` (`disabled: true`).
- **Line breaks:** line resolver counts only `\n`; `\r\n` on Windows still works but `\r` remains in text.
- **Entity decoding:** requires semicolon (`©` yes, `&copy` no) — acceptable, just document it.
- **`showExample1/2` still output placeholder JSON**, not real `html2json()` output (demo/submission polish issue).

What is already strong

- No DOM parser (requirement met)
- Good token priority (raw elements before generic tags)
- Quote-aware raw closing for script/style (important edge case)
- Void/self-closing handling
- Warning model with `openedAt/detectedAt` is clear and useful
- Always returns stable root shape with `warnings`
- Parses original input (good fix for line accuracy)
- Test suite is broad (47 tests + samples)

Submission advice (practical)

Before submit, I'd do only these high-value checks:

1. Align schema comment with actual whitespace behavior.
2. Manually test 3 warning cases in `index.html`:
   - unexpected `</span>`
   - unclosed `<div>`
   - unclosed `<!-- comment`
3. Update `showExample1/2` to call `html2json()` (better demo).
4. Complete `ai_help` link file.
5. Be ready to explain trade-offs: best-effort recovery vs strict HTML5 compliance.

If you want, I can give you a short “interview answer sheet” (design decisions + known limitations) based on your current implementation.

User

Let's fix Minor inconsistencies step-by-step

Cursor

Fixing minor inconsistencies step-by-step. Reviewing the current implementation and related files.

Fixing minor inconsistencies step-by-step: updating the schema comment first, then line-break handling, then wiring the demo examples.

Adding a test for CRLF line-number handling, then running the suite.

All four minor inconsistencies are fixed. Here’s what changed:

Step 1 — Schema comment aligned with behavior
- **Whitespace:** now documents that whitespace is preserved (matches tests and code).
- **Attributes:** typed as `string | boolean` (boolean attrs like `disabled: true`).
- **Entities:** documented that a trailing semicolon is required (`©`, not `&copy`).

Step 2 — Line resolver handles Windows line endings
`buildLineStartIndexes` now treats `\r\n` as one break and standalone `\r` as a break, so warning line numbers are correct on CRLF input.

Step 3 — Demo examples use real conversion
`showExample1` and `showExample2` now call `convertHtml2JsonAndSet()` instead of showing placeholder JSON.

Step 4 — Test coverage
Added `counts CRLF line breaks in warning line numbers` — **48/48 tests pass**.

Note: `\r` characters can still appear inside text node `content` on CRLF input; only line *numbering* was fixed. Normalizing `\r\n` → `\n` in text would be a separate change if you want that next.

The **real logic issues** from the earlier review (unexpected-close `openedAt`, EOF `detectedAt`, unclosed comment behavior) are still open — say if you want to tackle those next.

User

I skipped showExample1 and showExample2 because they were provided. Let's work on Normalizing \r\n → \n in text

---

Cursor

Implementing `\r\n` → `\n` normalization in text nodes. Reviewing where text content is produced in the parser.

Adding a single normalization helper at text-node creation and updating the schema comment.

Line ending normalization is in place. Summary:

Implementation

Added `normalizeTextLineEndings()` using `/\r\n?/g → "\n"`, applied when text nodes are created:

- **Regular text** — in `createTextNode` (before entity decoding)
- **Script/style raw text** — in `createRawTextNode` (entities stay literal; only line endings change)
- **Textarea/title** — via `createTextNode` (normalize + decode entities)

Not normalized (by design): comments, doctype, attribute values.

Schema

Documented the policy: text nodes normalize `\r\n` and standalone `\r` to `\n`.

Tests (4 new, 52 total passing)

- CRLF in plain text
- Standalone `\r` in plain text
- CRLF in script (entities unchanged)
- CRLF in textarea (entities decoded)

Warning line numbers still use the original input indices via `buildLineStartIndexes` — only `content` in text nodes is normalized.

User

"The real logic issues from the earlier review (unexpected-close openedAt, EOF detectedAt, unclosed comment behavior) are still open — say if you want to tackle those next." - what do you mean?

Cursor

Reviewing the warning logic in your parser so I can explain those three issues precisely.

Those three items were **optional polish points** about warning semantics and malformed-input recovery — not crashes or test failures. Your parser works; these are places where behavior can feel inconsistent or confusing.

---

1. Unexpected closing tag — confusing `openedAt`

When `</span>` has no matching `<span>` on the stack:

```484:491:html2json.js
  if (foundTagIndex === -1) {
    warnings.push(
      createWarning(
        `Unexpected closing tag </${tag}>. No matching opening tag was found.`,
        getOpenLine(stack[stack.length - 1]),
        tokenLine
      )
    );
```

For input like:

```html
<div>
  </span>
</div>
```

You get:
- `openedAt: 1` → line where **`<div>`** opened (current parent)
- `detectedAt: 2` → line where **`</span>`** appeared

That matches your test, but **`openedAt` sounds like “where did the bad tag open?”** — and `</span>` never opened. A reader might think `<span>` opened on line 1.

What “fix” would mean: use `detectedAt` for both, omit `openedAt`, or rename/clarify the field for this warning type.

---

2. Unclosed tag at EOF — `detectedAt` points to opening line

At end of input, unclosed tags get:

```604:612:html2json.js
  while (stack.length > 1) {
    const unclosedElement = stack.pop();
    warnings.push(
      createWarning(
        `Unclosed tag <${unclosedElement.tag}> was automatically closed at the end of input.`,
        getOpenLine(unclosedElement),
        getOpenLine(unclosedElement)
      )
    );
```

Both `openedAt` and `detectedAt` are the **opening line** of the tag.

For a long file where `<div>` opens on line 5 and EOF is line 200, you’d expect:
- `openedAt: 5` — where `<div>` opened
- `detectedAt: 200` — where the problem was noticed (EOF)

Right now both are `5`, so it’s harder to see *when* the missing close was detected.

---

3. Unclosed comment — warn, but don’t stay in “comment mode”

For:

```html
<div>
<!-- comment
<p>Text</p>
```

Flow:
1. `tryReadComment` finds `<!--` but no `-->` → returns `null`
2. Parser emits **one** warning (line 576–584)
3. It treats the first `<` as literal text and continues normal parsing
4. `<p>Text</p>` becomes a **real element**, not comment content

So the warning says *“comment is not closed”*, but the tree still parses HTML after it as if the comment never started. That’s **best-effort recovery**, not a full comment state machine.

What “fix” would mean: after `<!--` without `-->`, consume everything until EOF (or next `-->`) as comment/text and **don’t** parse inner tags — closer to browser behavior.

Bottom line

Issue	Severity	What it affects
Unexpected-close `openedAt`	UX / naming	Warning readability
EOF `detectedAt`	UX	Pinpointing where error was found
Unclosed comment	Recovery semantics	Tree shape for broken comments

None of these block submission. They’re trade-offs you can **keep and explain in the interview**, or **fix** if you want clearer diagnostics / stricter recovery.

If you want to tackle one, the EOF `detectedAt` fix is usually the smallest and most obviously useful change.

User

Check my current warnings logic, change tests if needed

Cursor

Reviewing the warnings implementation and tests to align behavior and expectations.

Fixing unexpected-close `openedAt` semantics and updating tests to match the EOF `detectedAt` behavior.

All **52 tests pass**. Here’s what I reviewed and changed.

Warning semantics (current logic)

Warning type	`openedAt`	`detectedAt`
Unexpected closing tag	Line of the stray `</tag>`	Same line (no opening tag exists)
Auto-close before mismatch	Line where inner tag opened	Line of the closing tag
Unclosed tag at EOF	Line where tag opened	**Last line of input** (`resolveLine(htmlText.length)`)
Unclosed comment	Line where `<!--` starts	Same line (detected immediately)
Unclosed raw tag (no `>`)	Line where tag starts	Same line

Code change

Unexpected closing tag — `openedAt` no longer points at the parent element (which was misleading). Both fields now use the closing tag’s line:

```484:491:html2json.js
  if (foundTagIndex === -1) {
    warnings.push(
```

```
    createWarning(
      `Unexpected closing tag </${tag}>. No matching opening tag was found.`,
      tokenLine,
      tokenLine
    )
  );
```

Your **EOF `detectedAt` fix** was already correct — I kept it as-is.

Test updates

- Unexpected close: `openedAt: 2, detectedAt: 2` (was `openedAt: 1`)
- CRLF unexpected close: same fix
- Unclosed comment + unclosed div: div `detectedAt: 3` (EOF after `</p>`, was `1`)

Schema comment

Added brief docs on what `openedAt` vs `detectedAt` mean.

Still intentional (not changed): unclosed `<!--` emits a warning but does not enter comment mode — content after it is still parsed as HTML. That's a recovery trade-off, not a warning-line bug. Say if you want that behavior changed next.

User

Write consise parser contract summary (just few sentences how it works) as a text (without tables), I want it add to README

Cursor

Here's a concise block you can paste into your README:

Parser contract

`html2json(htmlText)` is a hand-written, character-by-character parser (no DOM APIs) that always returns a JSON AST rooted at `{ type: "document", children, warnings }`. The tree uses four node kinds: `element` (with lowercase `tag`, `attributes`, and `children`), `text`, `comment`, and `doctype`. Malformed HTML is recovered best-effort: the function never throws, and structural problems are reported in `warnings` with 1-based `openedAt` / `detectedAt` line numbers.

Text nodes preserve whitespace as written and normalize line endings (`\r\n` and `\r` → `\n`). HTML entities are decoded in text and attribute values (semicolon required), but stay literal inside `script`, `style`, comments, and doctype. Content inside `script`, `style`, `textarea`, and

`title` is treated as raw text (with quote-aware closing-tag detection for script/style). Void and self-closing tags are supported; unclosed or mismatched tags produce warnings and an approximate tree rather than failing.